

$$\begin{cases} \max & \int_0^{\infty} e^{-st} \log c_t dt \\ \text{s.t.} & \dot{B}_t = 2B_t + y_t - c_t \end{cases}$$

$$H_t = e^{-st} \log c_t + (2B_t + y_t - c_t) \cdot \lambda_t$$

$$\frac{\partial H_t}{\partial c_t} = e^{-st} \frac{1}{c_t} - \lambda_t = 0 \quad \lambda_t = e^{-st} \frac{1}{c_t}$$

$$\frac{\partial H_t}{\partial B_t} = 2\lambda_t = -\dot{\lambda}_t \quad \Rightarrow \quad \dot{\lambda}_t = -s \cdot e^{-st} \frac{1}{c_t} - e^{-st} \frac{\dot{c}_t}{c_t^2}$$

$$\lim_{t \rightarrow \infty} \lambda_t B_t = 0$$

$$2 e^{-st} \frac{1}{c_t} = s e^{-st} \frac{1}{c_t} + e^{-st} \frac{\dot{c}_t}{c_t^2}$$

$$\boxed{\frac{\dot{c}_t}{c_t} = 2 - s} \quad (EE)$$

$$\frac{dc_t}{c_t} = (2-s) dt$$

$$\int_0^t \frac{dc_\tau}{c_\tau} = \int_0^t (2-s) d\tau \quad \Rightarrow \quad \log c_\tau \Big|_0^t = (2-s) \tau \Big|_0^t$$

$$\Rightarrow \underbrace{\log c_t - \log c_0}_{\log(c_t/c_0)} = (2-s)(t-0)$$

$$\Rightarrow \frac{c_t}{c_0} = e^{(2-s)t}$$

$$\Rightarrow \boxed{c_t = c_0 \cdot e^{(2-s)t}}$$

$$e^{-2t} \dot{B}_t - e^{-2t} \cdot 2B_t = e^{-2t} (y_t - c_t)$$

$$\frac{d(e^{-2t} B_t)}{dt}$$

$$\Rightarrow \int_0^{\infty} d(e^{-2t} B_t) = \int_0^{\infty} e^{-2t} (y_t - c_t) dt$$

$$e^{-2t} B_t \Big|_0^{\infty} = \left(-\frac{1}{2} e^{-2t} y + \frac{1}{s} e^{-st} c_0 \right) \Big|_0^{\infty} = \int_0^{\infty} (e^{-2t} y - e^{-st} c_0) dt$$

$$\dot{X}_t = 2 \Rightarrow \frac{dX_t}{dt} = 2 \Rightarrow \int dX_t = \int 2 dt \Rightarrow X_t = 2t + c$$

$$\lim_{t \rightarrow \infty} e^{-2t} B_t - B_0 = \frac{1}{2} y - \frac{1}{s} c_0 \Rightarrow \boxed{c_0 = s \left[B_0 + \frac{1}{2} y \right]}$$

$$\text{TRC: } \lim_{t \rightarrow \infty} e^{-st} \frac{B_t}{c_t} = \lim_{t \rightarrow \infty} e^{-st} \frac{B_t}{c_0 e^{(2-s)t}} = \lim_{t \rightarrow \infty} \frac{e^{-2t} B_t}{c_0} = 0$$

$$F(\vec{x}_t, \vec{x}_{t+1}, \vec{x}_{t-1}, \dots) = 0$$

Dynamic System

Linear

$$x_{t+1} = A x_t$$

Non-linear

1st order

High-order

$$x_{t+1}(x_t, x_{t-1}, x_{t-2})$$

1 eq. $N > 1$ eq.

① Linear dynamic equation

$$x_{t+1} = \lambda \cdot x_t \Rightarrow x_t = \lambda x_{t-1} = \lambda (\lambda x_{t-2}) = \dots = \lambda^t \cdot x_0$$

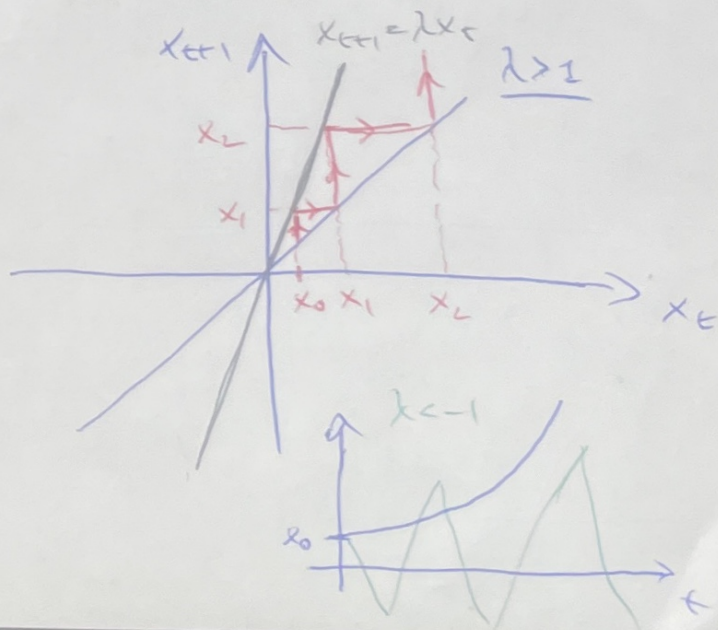
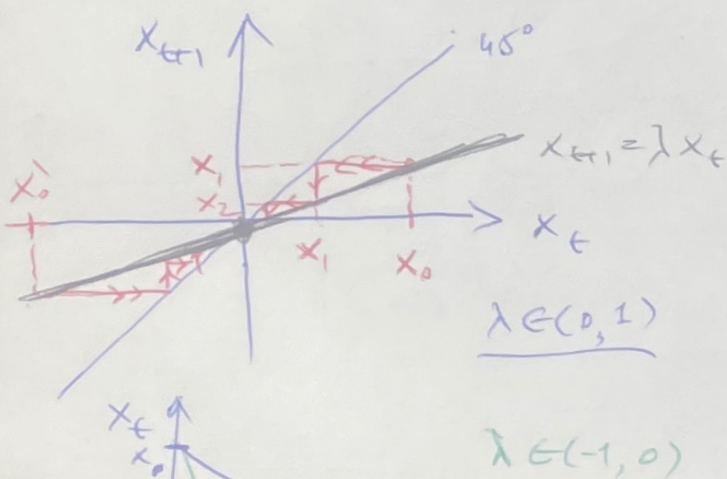
Steady state $\bar{x}$: if $x_t = \bar{x}$, then $x_{t+1} = \bar{x}$
 $\bar{x} = \lambda \cdot \bar{x} \Rightarrow (1-\lambda) \bar{x} = 0$
 $\lambda x_t = \lambda \bar{x}$

if $\lambda = 1 \Rightarrow \forall \bar{x} \in (-\infty, +\infty)$ is SS

if $\lambda \neq 1 \Rightarrow \bar{x} = 0$ is unique SS

$\bar{x}$ is stable if $x_t \rightarrow \bar{x}$ as $t \rightarrow \infty$

$x_t = \lambda^t \cdot x_0$ $\bar{x} = 0$ is stable if $|\lambda| < 1$



$$x_{t+1} = \lambda_1 x_t + \lambda_2 x_{t-1}$$

$$\begin{pmatrix} x_{t+1} \\ x_t \end{pmatrix} = \begin{pmatrix} \lambda_1 & \lambda_2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x_t \\ x_{t-1} \end{pmatrix}$$

$$\underbrace{\begin{pmatrix} x_{t+1} \\ x_t \end{pmatrix}}_{z_{t+1}} = \underbrace{\begin{pmatrix} \lambda_1 & \lambda_2 \\ 1 & 0 \end{pmatrix}}_A \underbrace{\begin{pmatrix} x_t \\ x_{t-1} \end{pmatrix}}_{z_t}$$

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}$$

$$x_{t+1} = A x_t, \quad x_t = \begin{pmatrix} x_{1t} \\ x_{2t} \end{pmatrix} \quad A_{2 \times 2}$$

$$\underline{SS}: x_t = x_{t+1} = \bar{x} \Rightarrow \bar{x} = A \bar{x} \Rightarrow \underbrace{(I - A)}_{\text{rank}(A-I)=2} \bar{x} = 0$$

$$\Rightarrow \text{rank}(A - I) = 2 \Rightarrow \bar{x} = 0$$

$$\Rightarrow \text{rank}(A - I) < 2 \Rightarrow \text{continuum of SS}$$

$$x_{t+1} = A x_t = A(A x_{t-1}) \Rightarrow$$

$$x_t = A^t \cdot x_0$$

$$\Rightarrow A = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} \Rightarrow$$

$$\begin{cases} x_{1t} = \lambda_1^t x_{10} \\ x_{2t} = \lambda_2^t x_{20} \end{cases}$$

$$A = Q \Lambda Q^{-1} \Rightarrow A Q = Q \Lambda, \quad Q = (v^1 \ v^2)$$

$$\Rightarrow A \cdot v^i = \lambda_i v^i$$

$$x_t = A^t x_0 = \underbrace{A}_{Q \Lambda Q^{-1}} \cdot \underbrace{A}_{Q \Lambda Q^{-1}} \cdots \underbrace{A}_{Q \Lambda Q^{-1}} \cdot x_0 = Q \underbrace{\Lambda^t}_{\begin{pmatrix} \lambda_1^t & 0 \\ 0 & \lambda_2^t \end{pmatrix}} Q^{-1} x_0$$

$$x_{t+1} = A x_t \Rightarrow x_{t+1} = Q \Lambda Q^{-1} x_t \Rightarrow \underbrace{Q^{-1} x_{t+1}}_{y_{t+1}} = \underbrace{\Lambda Q^{-1} x_t}_{y_t}$$

$$\Rightarrow y_{t+1} = \Lambda y_t \Rightarrow \begin{cases} y_{1t} = \lambda_1^t y_{10} \\ y_{2t} = \lambda_2^t y_{20} \end{cases}$$

$$A = \begin{pmatrix} 1 & 0.5 \\ 1 & 1.5 \end{pmatrix}$$

(4)

$$1) |A - \lambda I| = \begin{vmatrix} 1-\lambda & 0.5 \\ 1 & 1.5-\lambda \end{vmatrix} = (1-\lambda)(1.5-\lambda) - 0.5$$

$$= \lambda^2 - 2.5\lambda + 1.5 - 0.5 = \lambda^2 - 2.5\lambda + 1 = 0$$

$$\lambda_1 = 2 \quad \lambda_2 = 0.5$$

$$\Lambda = \begin{pmatrix} 2 & 0 \\ 0 & 0.5 \end{pmatrix}$$

$$2) (A - \lambda_1 I) v^1 = 0 \Rightarrow \begin{pmatrix} -1 & 0.5 \\ 1 & -0.5 \end{pmatrix} v^1 = 0$$

$$v^1 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

λ_2

$$(A - \lambda_2 I) v^2 = 0 \Rightarrow \begin{pmatrix} 0.5 & 0.5 \\ 1 & 1 \end{pmatrix} v^2 = 0$$

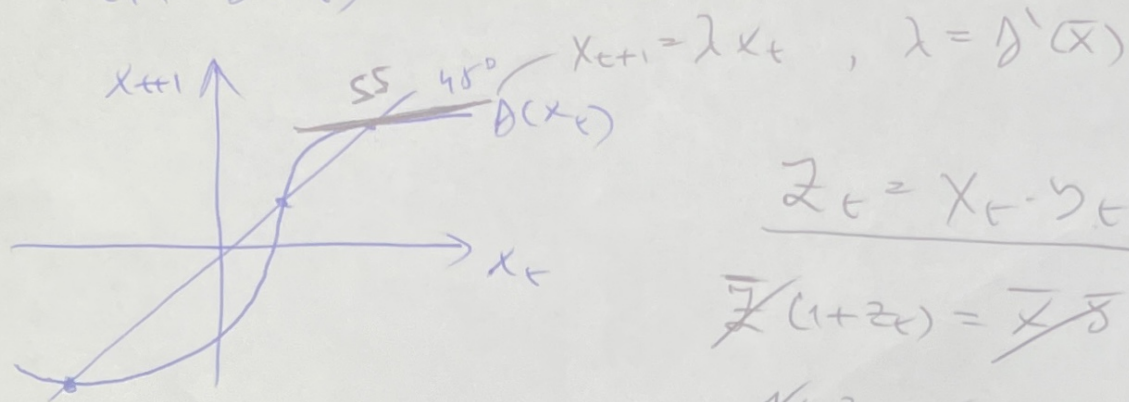
$$v^2 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$3) Q = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} \quad Q^{-1} = \frac{1}{-1-2} \begin{pmatrix} -1 & -1 \\ -2 & 1 \end{pmatrix} = \frac{1}{3} \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix}$$

$$4) x_t = Q \Lambda^t \cdot Q^{-1} x_0 = \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} 2^t & 0 \\ 0 & 0.5^t \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 2 & -1 \end{pmatrix} \cdot \frac{1}{3} \cdot \begin{pmatrix} 2 \\ 1 \end{pmatrix}$$

$$x_{t+1} = \delta(x_t)$$

(6)



$$z_t = x_t - y_t$$

$$\cancel{\bar{x}}(1+z_t) = \cancel{\bar{x}}\delta \cdot (1+x_t)(1+y_t)$$

$$x + z_t = x + x_t + y_t + \cancel{x_t y_t}$$

$$(z_t = x_t + y_t)$$

$$X(t) = \begin{pmatrix} x_1(t) \\ x_2(t) \end{pmatrix}$$